Erdős #377: the carrier criterion, the carry conservation law, and the exact location of the remaining difficulty

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Abstract

#377 is not proved here. What is proved: the balance condition is an exact statement about a single phasor on the carrier — n is p-balanced iff $\{n/p^i\} < \cos\theta_p$ for every $i \geq 1$, with $\cos\theta_p = \frac{p+1}{2p}$ — so that payoff and alignment cost are one angle read two ways, and the cost of freezing rail p at n is exactly $\beta_p \log n$. From Kummer's carry count we obtain an exact conservation law, $\sum_p \kappa_p(n)\log p = \log\binom{2n}{n}$, and from it an unconditional evaluation valid for *every* n: the θ-weighted mass of balanced primes above $\sqrt{2n}$ equals $2n(1-\log 2)+o(n)$. We then show this law is *orthogonal* to #377 — its support on small primes is $O(\sqrt{n})$ out of $2n\log 2$, measured share $0.0127 \to 0.00066$ — so it cannot bound E. Two candidate reductions are tested and one is refuted: the entropy budget $\sum_{p\in S(n)} \beta_p \leq 1$ is **false** for every fixed truncation (measured maximum 7.03), holding only as $d_p \to \infty$. What survives is a clean reduction with an explicit constant, and a difficulty localised to a single range of primes, stated in §6.

1 The carrier criterion

For an odd prime p let $D_p = \{0, \ldots, \frac{p-1}{2}\}$, and call n *p-balanced* ($n \in G_p$) if every base-p digit of n lies in D_p; by Kummer this is exactly $p \nmid \binom{2n}{n}$. Write

$$\cos\theta_p = \frac{|D_p|}{p} = \frac{p+1}{2p}, \qquad \theta_p \nearrow \frac{\pi}{3}, \qquad \frac{\pi}{3} - \theta_p = \frac{1}{\sqrt{3}\,p}\big(1 + O(p^{-1})\big).$$

Theorem 1 (carrier criterion: the universal half-turn)**.** *Let $F(x) = \mathbf{1}[\{x\} < \frac{1}{2}]$. For every $n \geq 1$ and every odd prime p,*

$$n \in G_p \iff n \bmod p^j < \tfrac{1}{2}p^j \ \ \forall j \geq 1 \iff F\Big(\frac{n}{p^j}\Big) = 1 \ \ \forall j \geq 1 \iff \Im\, e\Big(\frac{n}{p^j}\Big) > 0 \ \ \forall j \geq 1.$$

Proof. ($\Rightarrow$) If every digit is $\leq \frac{p-1}{2}$ then $n \bmod p^j = \sum_{i<j} n_i p^i \leq \frac{p-1}{2}\cdot\frac{p^j-1}{p-1} = \frac{p^j-1}{2} < \frac{p^j}{2}$. ($\Leftarrow$) Induct on j. From $n \bmod p < \frac{p}{2}$ we get $n_0 \leq \frac{p-1}{2}$. Given $n_0, \ldots, n_{j-1} \leq \frac{p-1}{2}$, the hypothesis $n \bmod p^{j+1} \leq \frac{p^{j+1}-1}{2}$ gives $n_j p^j \leq \frac{p^{j+1}-1}{2}$, so $n_j \leq \frac{p^{j+1}-1}{2p^j} < \frac{p}{2}$, i.e. $n_j \leq \frac{p-1}{2}$. The last two equivalences are $\{x\} < \frac{1}{2} \iff \sin 2\pi x > 0$. □

Verified: 480 000 pairs (n, p), $p \in \{3, \ldots, 1009\}$, zero mismatches against the digit criterion.

The condition has no digits in it and, more to the point, *no per-rail constant*: one universal function F, threshold exactly the half-turn, applied at the scales p^j. The rails differ only in their multiplier. Since $F = \frac{1}{2} + \frac{2}{\pi}\sum_{k\ \mathrm{odd}} \frac{\sin 2\pi k x}{k}$ carries odd harmonics only, each with $1/k$ decay, the second (even, non-decaying) lane seen in the residue-wise reading is the aliasing of that tail onto $\mathbb{Z}/p$, not a feature of the carrier — though it is *not* removed by the reformulation: the two readings give literally the same subset of $\mathbb{Z}/p$, hence identical Fourier coefficients, checked to eight places.

Writing $\cos\theta_p = \frac{p+1}{2p}$ for the density of D_p (so $\theta_p \nearrow \frac{\pi}{3}$ and $\cos\theta_p - \cos\frac{\pi}{3} = \frac{1}{2p}$ exactly), payoff and cost are one angle:

$$\frac{1}{p} = \frac{2}{p}\Big(\cos\theta_p - \cos\tfrac{\pi}{3}\Big)\cdot 1, \qquad \beta_p := 1 - \frac{\log|D_p|}{\log p} = \frac{\log\sec\theta_p}{\log p}.$$

Proposition 2 (cost identity)**.** *Let $d_p = \lfloor \log n/\log p\rfloor$. The density of G_p among the residues mod p^{d_p+1} is $\cos^{d_p+1}\theta_p$, so the cost of freezing rail p at height n is*

$$-\log\big(\mathit{density}\big) = (d_p+1)\log\sec\theta_p = \beta_p \log n + O(\log p).$$

Write $S(n) = \{p \le n : \ n \in G_p\}$ and $E(n) = \sum_{p\in S(n)} 1/p = \mathrm{pay}(S(n))$.

2 Band truncation (unconditional)

Theorem 3. *For every n and every integer $K \ge 1$,*

$$E(n) \ \le\ \log(K+1) \ +\ \sum_{p\le n^{1/K}} \frac{1}{p}\mathbf{1}[n\in G_p] \ +\ O\Big(\frac{1}{\log n}\Big).$$

Proof. Bound band $k = \{p \in (n^{1/(k+1)}, n^{1/k}]\}$ for $k < K$ by its full Mertens mass $\log\frac{k+1}{k} + O(e^{-c\sqrt{\log n}})$; the sum telescopes to $\log K$, and the band containing $n^{1/K}$ contributes at most $\log\frac{K+1}{K}$. □

No equidistribution enters, and no band is discarded. The whole of #377 is therefore the deep sum $\sum_{p\le n^{1/K}} \frac{1}{p}\mathbf{1}[n\in G_p]$.

3 What a budget would buy

Lemma 4 (payoff/cost monotonicity)**.** *With $\beta(x) = \log\frac{2x}{x+1}/\log x$, the ratio $r(x) = \frac{1/x}{\beta(x)} = \frac{\log x}{x\log\frac{2x}{x+1}}$ is strictly decreasing on (e,∞) — in particular on the odd primes — and $r(x) \sim \frac{\log x}{x\log 2} \to 0$.*

Proof. Write $r = uv$ with $u(x) = \frac{\log x}{x}$ and $v(x) = 1/g(x)$, where $g(x) = \log\frac{2x}{x+1} = \log 2 - \log(1+\frac{1}{x})$. Both are positive for $x > 1$, since $1+\frac{1}{x} < 2$ gives $g > 0$. Now $u'(x) = (1-\log x)/x^2 < 0$ for $x > e$, so u strictly decreases there; and $x \mapsto \log(1+\frac{1}{x})$ strictly decreases, so g strictly increases and $v = 1/g$ strictly decreases. A product of two positive strictly decreasing functions strictly decreases. The asymptotic follows from $g(x) \to \log 2$. □

Theorem 5 (LP reduction). *Let $\Lambda(B) = \sup\{\mathrm{pay}(S) : S$ a finite set of odd primes, $\sum_{p\in S}\beta_p \le B\}$. If for some K and some $B < \infty$*

$$\sum_{p\in S(n),\ p\le n^{1/K}} \beta_p \ \le\ B \qquad \text{for all sufficiently large } n, \tag{$\mathrm{B}_{K,B}$}$$

then $\limsup_n E(n) \le \log(K+1) + \Lambda(B)$; in particular #377 holds. Because $\frac{1/p}{\beta_p} = \frac{\log p}{p\log\frac{2p}{p+1}}$ is strictly decreasing, Λ is computed greedily; the integral values are $\Lambda(1) = 0.676190$ (rails $3, 5, 7$), $\Lambda(2.5) = 1.033439$ (rails up to 29), $\Lambda(7) = 1.417404$ (rails up to 173), against the fractional relaxation $\Lambda^{\mathrm{LP}}(1) = 0.685523$.

Proof. The bound $\limsup E \le \log(K+1) + \Lambda(B)$ is immediate from Theorem 3 and the definition of Λ. For the greedy evaluation, consider the fractional relaxation

$$\Lambda^{\mathrm{LP}}(B) = \max\Big\{\sum_p \frac{x_p}{p} \ :\ \sum_p x_p\beta_p \le B,\ x_p \in [0,1]\Big\} \ \ge\ \Lambda(B),$$

and write $r(p) = \frac{1/p}{\beta_p}$, strictly decreasing by Lemma 4. Let x be feasible and suppose it is not greedy: there are primes $p < q$ with $x_p < 1$ and $x_q > 0$. For small $\delta > 0$ set $x'_q = x_q - \delta$ and $x'_p = x_p + \delta\beta_q/\beta_p$, leaving all other coordinates fixed. This is again feasible (the budget used is unchanged, and $x'_p \le 1$, $x'_q \ge 0$ for δ small), and the objective changes by

$$\frac{\delta\beta_q}{\beta_p}\cdot\frac{1}{p} - \frac{\delta}{q} \ =\ \delta\beta_q\Big(\frac{1/p}{\beta_p} - \frac{1/q}{\beta_q}\Big) \ =\ \delta\beta_q\big(r(p) - r(q)\big) \ >\ 0,$$

strictly, since r is strictly decreasing and $\beta_q > 0$. Hence no non-greedy point is optimal, so the optimum fills the coordinates in increasing order of p until the budget is exhausted — which is the greedy rule. The integral values follow by evaluating the greedy solution and rounding down at the last prime. □

Theorem 6 ($\mathrm{B}_{K,1}$ is false). *For every fixed K, $(\mathrm{B}_{K,1})$ fails. Over the dyadic window $[N/2, N]$ the maximum of $\sum_{p\le z,\ n\in G_p}\beta_p$ is*

	$z=7$	$z=31$	$z=101$	$z=1009$
$N=10^6$	0.686	1.261	1.879	6.803
$N=10^7$	0.657	1.133	1.735	7.031
$N=4\times10^7$	0.686	1.283	1.637	6.575

and the 90th percentile at $z = 1009$ is already 4.06–5.05.

The reason is structural and worth stating, because it says where the budget *does* live. The cost identity (Proposition 2) charges $\beta_p \log n$ per rail only when the rail is asked to freeze at every one of its $d_p + 1$ scales. For p near $n^{1/K}$ with K fixed, $d_p = K$ is bounded, the freeze is cheap, and the number of such rails is large: their costs add without any of them being individually improbable. The budget binds only as $d_p \to \infty$, i.e. for $p \le n^{o(1)}$. Correspondingly the exceptional set at fixed z is *finite*: at $z = 31$ the count of n in the window with cost > 1 is $44,\ 39,\ 28$ while the window grows by a factor 40.

4 Band one is an exact union of intervals

Above $\sqrt{n}$ the balanced condition is not merely an interval union in principle: the endpoints are explicit rational functions of n and $m = \lfloor n/p \rfloor$.

Theorem 7 (exact interval criterion). *Let $p > \sqrt{n}$ be prime and put $m = \lfloor n/p \rfloor$. Then*

$$n \in G_p \qquad \Longleftrightarrow \qquad p\,(2m+1) \;\geq\; 2n+1 \quad \textit{and} \quad p \;\geq\; 2m+1.$$

Proof. Since $p > \sqrt{n}$ we have $p^2 > n$, so n has exactly the two base-p digits $n = mp + r$ with $r = n - mp$ and $0 \leq m < p$. By Kummer, $n \in G_p$ iff both digits are at most $\frac{p-1}{2}$. The bottom digit condition $r \leq \frac{p-1}{2}$ reads $2(n - mp) \leq p - 1$, i.e. $2n + 1 \leq p(2m + 1)$. The top digit condition $m \leq \frac{p-1}{2}$ reads $p \geq 2m + 1$. □

Verified: 1 134 661 pairs (n, p) with $n < 4000$ and $p > \sqrt{n}$ — no mismatch.

Corollary 8 (the band-one mass, exactly). *For each $m \geq 1$ the balanced primes p with $\lfloor n/p \rfloor = m$ are exactly the primes of*

$$I_m(n) \;=\; \Big[\;\max\!\Big(\tfrac{2n+1}{2m+1},\, 2m+1\Big),\; \tfrac{n}{m}\;\Big], \qquad \textit{so} \qquad \sum_{\substack{\sqrt{n}<p\leq n\\ n\in G_p}} \frac{1}{p} \;=\; \sum_{m\geq 1}\sum_{p\in I_m(n)} \frac{1}{p}.$$

Proof. Immediate from Theorem 7: $\lfloor n/p \rfloor = m$ is equivalent to $\frac{n}{m+1} < p \leq \frac{n}{m}$, and the first condition cuts that range at $\frac{2n+1}{2m+1}$, which lies inside it. □

Verified: the interval formula reproduces the direct sum to machine precision at $n = 500, 1000, 2000, 4000$, giving 0.273946, 0.291506, 0.337442, 0.301129.

Remark 9. *The endpoints $\frac{n}{m}$ and $\frac{2n+1}{2m+1}$ are consecutive mediants, so the balanced fraction of the m-th interval is $\log \frac{\log(n/m)}{\log((2n+1)/(2m+1))}$, which tends to $\frac{1}{2}\log\frac{m+1}{m}$ for fixed m as $n \to \infty$: exactly half the band, because the mediant bisects the interval logarithmically. The top-digit condition $p \geq 2m + 1$ binds only for $m \gtrsim \sqrt{n/2}$, i.e. only at the very bottom of the band.*

5 The carry conservation law

Kummer's criterion in its quantitative form is $v_p\binom{2n}{n} = \kappa_p(n)$, the number of carries in the base-p addition $n + n$; equivalently $\kappa_p(n) = \frac{2s_p(n) - s_p(2n)}{p-1} \geq 0$, and $n \in G_p \iff \kappa_p(n) = 0$. Unlike the indicator, κ_p is *one-signed and additive*, and it obeys an exact global identity.

Theorem 10 (conservation). *For every $n \geq 1$,*

$$\sum_{p\leq 2n} \kappa_p(n)\log p = \log\binom{2n}{n}.$$

Consequently, for every n,

$$\sum_{\substack{\sqrt{2n}<p\leq 2n\\ p\nmid\binom{2n}{n}}} \log p \;=\; 2n\,(1-\log 2) \;+\; O\!\big(\sqrt{n}\,\log n\big) \;=\; 0.613706\ldots\, n \;+\; o(n).$$

Proof. The identity is Legendre's formula summed over p. For $p > \sqrt{2n}$ the addition has at most two digits, so $\kappa_p \in \{0,1\}$; for $p \le \sqrt{2n}$, $\kappa_p \le \log(2n)/\log p$, whence $\sum_{p\le\sqrt{2n}} \kappa_p \log p \le \psi(\sqrt{2n}) = O(\sqrt{n})$. Therefore $\sum_{\sqrt{2n}<p\le 2n} \mathbf{1}[\kappa_p \ge 1]\log p = \log\binom{2n}{n} + O(\sqrt{n}) = 2n\log 2 + O(\sqrt{n}\log n)$, and subtracting from $\theta(2n) - \theta(\sqrt{2n}) = 2n + o(n)$ gives the display. □

Measured, ratio of the left side to $2n(1-\log 2)$ at $n = 10^4, 10^5, 10^6, 3\times 10^6$: 0.9771, 0.9939, 0.9979, 0.9985.

This is a genuine common-mode evaluation: exact, uniform in n, and free of any equidistribution input. It also settles the question it raises.

Proposition 11 (the law is orthogonal to #377)**.** *The share of* $\log\binom{2n}{n}$ *carried by primes* $p \le \sqrt{2n}$ *is* $O(\sqrt{n}/n) \to 0$ *— measured* 0.01268, 0.00373, 0.00106, 0.00066 *at* $n = 10^4, 10^5, 10^6, 3\times 10^6$*. Hence Theorem 10 constrains only the range* $p > \sqrt{2n}$*, where by Theorem 3 the Mertens mass is already at most* $\log 2$ *unconditionally. It gives* no *information about* $p \le n^{1/K}$*, which by Theorem 3 is the whole problem.*

The law is stated and proved here rather than suppressed, because a conservation identity of exactly this shape is what the method calls for, and knowing that it is inert is worth more than not having looked: the $\log p$ weight of the identity and the $1/p$ weight of E are dual, and the identity's mass sits where E's does not.

6 Falsification register

Pre-committed, and all currently *unhit*: (i) an n with $E_{\le y}(n) > \frac{1}{3} + \frac{1}{5} + \frac{1}{7}$ for $y < 3.98\times 10^{11}$ and infinitely many such n; (ii) a failure of Theorem 10's ratio to approach 1; (iii) a fixed z at which $\#\{n \le N : \mathrm{cost}_{\le z}(n) > 1\}$ grows with N. Hit and retracted this session: the entropy budget ($\mathrm{B}_{K,1}$), refuted by Theorem 6; and an earlier maximum of E attained at $n = 0$, which is balanced at every rail and is an artifact of an unrestricted range, not a wall.

7 Where the difficulty is, exactly

Let $\gamma_k(n)$ be the fraction of band k's Mertens mass that is balanced, so $E(n) = \sum_k \gamma_k(n)\log\frac{k+1}{k} + O(1/\log n)$. Band k imposes $k+1$ conditions, of which the top one is free for all but a $O(1/\log n)$ fraction of the band (Theorem 1 with $i = k+1$ reads $n/p^{k+1} < \cos\theta_p$, one cut). Hence the band common mode is

$$\mathbb{E}\,\gamma_k = \cos^k\theta_p + o(1) = 2^{-k} + o(1), \qquad \sum_{k\ge 1} 2^{-k}\log\frac{k+1}{k} = C_0 = 0.507834\ldots,$$

which is the mean of E — the band decomposition and the global common mode agree exactly. *Measured* at $N = 10^6$ over 1500 sampled $n \in [N/2, N]$:

k	$\#p$	band mass	mean γ_k	2^{-k}	max γ_k
1	78330	0.68925	0.47664	0.5	0.49747
2	143	0.39526	0.22686	0.25	0.36799
3	14	0.23712	0.13231	0.125	0.44554
4	5	0.22167	0.06347	0.0625	0.58912

Band 1 is concentrated to ± 0.02 over the whole sample; $\max_n \gamma_k$ rises toward 1 only as the band's prime count falls. No sampled n raised more than four of five bands above twice their common mode, and $\max_n \sum_{k\le 6} \gamma_k \log \frac{k+1}{k} = 0.6806$ against the trivial 1.7111. **Boundedness of E is therefore a joint constraint across near-independent channels, not a per-band decay:** each γ_k can individually approach 1, and what fails is doing so at many k for one n.

The prime count of band k is $\asymp k\, n^{1/k}/\log n$, which is large exactly when $p \gtrsim \log n$. This splits #377 into two halves, and both are open:

$\boxed{\text{(a) } p \le \log n : \ \pi(\log n) \text{ rails, mass } \log\log\log n}$ $\boxed{\text{(b) } p > \log n : \ \text{concentration of } \gamma_k}$

Half (a) is where the budget binds ($d_p \ge \log n/\log\log n$), and the ceiling of the companion note caps it at $\Lambda(1) = 0.676190$ — conditional on transversality for subsets of $\{p \le \log n\}$, a growing family. Half (b) needs $\gamma_k(n) \ll 1/(\log k)^{1+\delta}$ uniformly in n, which is far weaker than the truth 2^{-k} but still an equidistribution statement: for fixed n, the points $\big(\{n/p\}, \ldots, \{n/p^k\}\big)$ must equidistribute in $[0,1)^k$ as p ranges over the band. Condition i has period $t^{i+1}/(in)$ in t, which exceeds 1 only for $i \ge k-1$; so at most two of the k conditions are resolvable at the integer scale, and the rest are questions about primes in intervals shorter than unit length. Band 1 is the exception and is proved (companion note, on $m \le n^{(1-\varepsilon)/2}$), giving $\gamma_1 = \frac{1}{2} + O(\varepsilon)$ — one band out of infinitely many.

That is the gap, in both halves, named and not papered over. Nothing in this note assumes either, and nothing in §1–§5 depends on them.

Remark 12 (register)**.** ***Proven unconditionally****: Theorems 1, 7, 3, 5 (as an implication), 10; Corollary 8; Propositions 2, 11. Theorem 7 and Corollary 8 give band one exactly: the balanced primes above $\sqrt{n}$ are precisely the primes of the explicit intervals $I_m(n)$, so the "free half" of band one is computed rather than estimated.* ***Refuted:*** $(\mathrm{B}_{K,1})$*, by measurement.* ***Not proved****: Erdős #377.* ***The literature gate has not been run****: Erdős–Graham–Ruzsa–Straus (1975) and successors must be read at source before any novelty claim; Theorem 10 in particular is an immediate consequence of Legendre's formula and should be presumed known.*